Christopher Tessum

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Professional Appointments

Assistant Professor—University of Illinois at Urbana-Champaign 2020–present
Department of Civil and Environmental Engineering

Research Scientist—University of Washington 2016–2019
Department of Civil and Environmental Engineering

Postdoctoral Associate—University of Minnesota 2015–2016
Department of Bioproducts and Biosystems Engineering

Education

Ph.D., Civil, Environmental, and Geo- Engineering (public health minor)—University of Minnesota 2009–2014

B.M.E., Mechanical Engineering (*cum laude*)—University of Minnesota 2002–2006

Honors and Awards

EPA Science to Achieve Results (STAR) Early Career Award 2020

NASA Early Career Faculty Award 2021

NSF CAREER Award 2024

American Geophysical Union (AGU) GeoHealth Early Career Award 2025

Selected Peer-Reviewed Publications (*=corresponding author; self and advisees are underlined)

16. Guo, L., X. Yang, Z. Zheng, N. Riemer, and C.W. Tessum* (2024) Uncertainty Quantification in Reduced-Order Gas-Phase Atmospheric Chemistry Modeling Using Ensemble SINDy. *Journal of Geophysical Research: Machine Learning and Computation*. **1**:4 e2024JH000358.

15. Yang, X., L. Guo, Z. Zheng, N. Riemer, and C.W. Tessum* (2024) Atmospheric Chemistry Surrogate Modeling With Sparse Identification of Nonlinear Dynamics. *Journal of Geophysical Research: Machine Learning and Computation*. **1**:2 e2024JH000132.

14. Park, M., Z. Zheng, N. Riemer, and C.W. Tessum* (2024) Learned 1D Passive Scalar Advection to Accelerate Chemical Transport Modeling: A Case Study with GEOS-FP Horizontal Wind Fields. *Artificial Intelligence for the Earth Systems*. **3**:3 e230080.

13. Giang, A., M.R. Edwards, S.M. Fletcher, R. Gardner-Frolick, R. Gryba, J. Mathias, C. Venier-Cambron, J.M. Anderies, E. Berglund, S. Carley, J.S. Erickson, E. Grubert, A. Hadjimichael, J. Hill, E. Mayfield, D. Nock, K.K. Pikok, R.K. Saari, M.S. Lezcano, A. Siddiqi, J.B. Skerker, and C.W. Tessum (2024) Equity and modeling in sustainability science: Examples and opportunities throughout the process. *Proceedings of the National Academy of Sciences*. **121**:13 e2215688121.

12. Wang, Y., J.S. Apte, J.D. Hill, C.E. Ivey, D. Johnson, E. Min, R. Morello-Frosch, R. Patterson, A.L. Robinson, C.W. Tessum, and J.D. Marshall (2023) Air quality policy should quantify effects on disparities. *Science*. **381**:6655 272-274.

11. Wang, Y., J.S. Apte, J.D. Hill, C.E. Ivey, R.F. Patterson, A.L. Robinson, C.W. Tessum, and J.D. Marshall (2022) Location-specific strategies for eliminating US national racial-ethnic PM2.5 exposure inequality. *Proc. Natl.*

Acad. Sci. U.S.A. **119**:44 e2205548119.

10. Domingo, N., S. Thakrar, S. Balasubramanian, K. Colgan, M. Clark, P. Adams, S. Pandis, D. Tilman, J. Marshall, C.W. Tessum, A.L. Goodkind, I.L. Azevedo, and J.D. Hill (2021) Air quality-related health damages of food. *Proc. Natl. Acad. Sci. U.S.A.* **118**:20 e2013637118.
9. Tessum, C.W.*, D.A. Paolella, S.E. Chambliss, J.S. Apte, J.D. Hill, and J.D. Marshall (2021) PM2.5 Polluters Disproportionately and Systemically Affect People of Color in the United States. *Science Adv*. **7**:18 eabf4491.
8. Thakrar, S.K., S. Balasubramanian, P.J. Adams, I. Azevedo, N.Z. Muller, S.N. Pandis, S. Polasky, P. Arden, A.L. Robinson, J.S. Apte, C.W. Tessum, J.D. Marshall, and J.D. Hill (2020) Reducing Mortality from Air Pollution in the United States by Targeting Specific Emission Sources. *Environ. Sci. Technol. Lett*. **7**:9 639–645.
7. Goodkind, A.L., C.W. Tessum, J.S. Coggins, J.D. Hill, and J.D. Marshall (2019) Fine-scale damage estimates of particulate matter air pollution reveal opportunities for location-specific mitigation of emissions. *Proc. Natl. Acad. Sci. U.S.A.* **116**:18 8775–8780.
6. Hill, J.D., A.L. Goodkind, C.W. Tessum, S. Thakrar, D. Tilman, S. Polasky, T. Smith, N. Hunt, K. Mullins, M. Clark, and J.D. Marshall (2019) Air-quality-related health damages of maize. *Nat. Sustain*. **2** 397–403.
5. Tessum, C.W., J.S. Apte, A.L. Goodkind, N.Z. Muller, K.A. Mullins, D.A. Paolella, S. Polasky, N.P. Springer, S.K. Thakrar, J.D. Marshall, and J.D. Hill (2019) Inequity in consumption of goods and services adds to racial-ethnic disparities in air pollution exposure. *Proc. Natl. Acad. Sci. U.S.A.* **116**:13 6001–6006 (featured article).
4. Tessum, C.W.*, J.D. Hill, and J.D. Marshall (2017) InMAP: A model for air pollution interventions. *PLoS ONE*. **12**:4 e0176131.
3. Tessum, C.W., J.D. Hill, and J.D. Marshall (2015) Twelve-month, 12 km resolution North American WRF-Chem v3.4 air quality simulation: performance evaluation. *Geosci. Model Dev*. **8**:4 957–973.
2. Tessum, C.W., J.D. Hill, and J.D. Marshall (2014) Life cycle air quality impacts of conventional and alternative light-duty transportation in the United States. *Proc. Natl. Acad. Sci. U.S.A.* **111**:52 18490–18495.
1. Tessum, C.W., J.D. Marshall, and J.D. Hill (2012) A spatially and temporally explicit life cycle inventory of air pollutants from gasoline and ethanol in the United States. *Environ. Sci. Technol*. **46**:20 11408–11417.

Student Research Advising

PhD Graduate: Amir Kazemi, Dissertation: "Advancing Generative AI for Enhanced Analytics in Urban and Environmental Monitoring" — 2023–2024

Current PhD Students: Shiyuan Wang, Xiaokai Yang, Lin Guo, Manho Park, Qurat ul ain Fatima, and Jialin Liu — 2020–present

Mentored five undergraduate students in REU programs, one high-school student researcher, and ten graduate student independent study projects, with one undergraduate serving a critical role in an externally funded research project — 2020–present